

Clustering of Spatial Data

An application to early childhood education and care services
and their impact on women's labour market

Tommaso Baresi, Luca Perego, Paul Poupeau,
Leonardo Tisato, Alexis Toppé, Anh Chi Tuan

February 19th, 2026



POLITECNICO
MILANO 1863

Introduction

Goal Detect spatial clusters that capture the joint variation in ECEC availability and female employment outcomes in Italian provinces

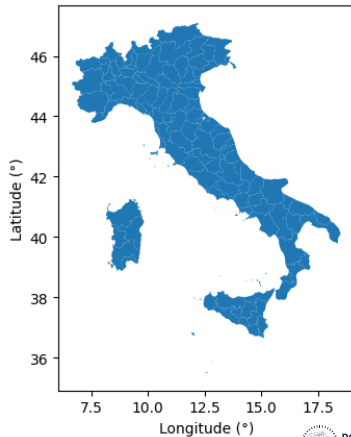
Steps to reach the goal

- Data cleaning and Exploratory Data Analysis
- Variable Selection
- Label Assignment

Dataset Overview

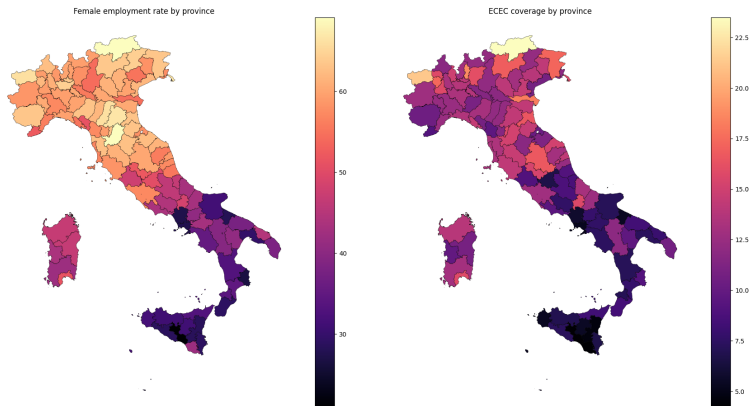
Analysis of a geojson file

- Data from 2022
- Main source: ISTAT
- 107 provinces
- 47 variables
- No missing values



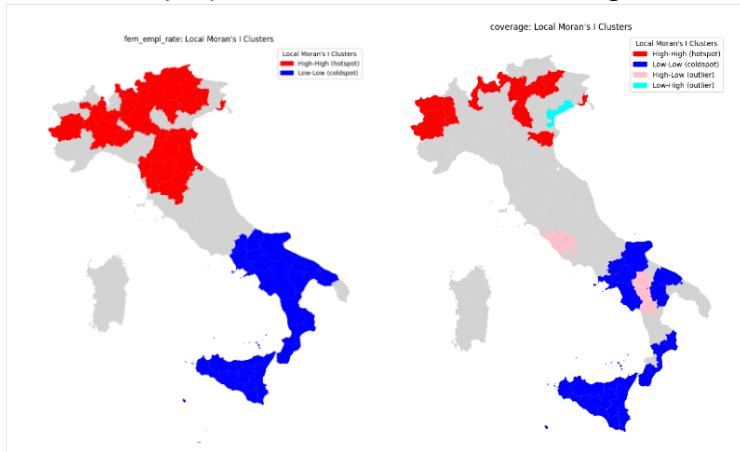
Spatial Visualization

Female employment rate and ECEC coverage by province



Moran's I Cluster

Female employment rate and ECEC coverage cluster



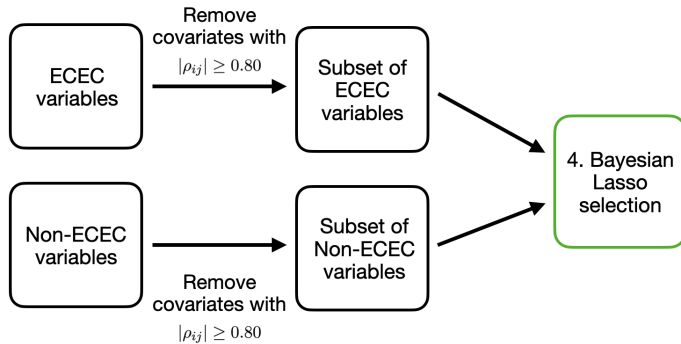
Variable Selection

Goal : identify the most relevant predictors.

1. Remove variables considered irrelevant by the sociological literature (ex : urban_type, area_km2, ...);
2. Remove variables highly correlated with the female employment rate ($|\rho| \geq 0.85$), as they partially convey the same information (ex : empl_rate, unempl_rate, fem_house_rate, ...)

Variable Selection

3. Mitigate multicollinearity inside the ECEC group and Non-ECEC group



Bayesian Lasso

$$y_i^* \mid \boldsymbol{\beta}, \sigma, \phi_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\alpha + \phi_i + \mathbf{x}_i^{*\top} \boldsymbol{\beta}, \sigma^2), \quad i = 1, \dots, N$$

$$\beta_j \mid \lambda, \sigma \stackrel{\text{iid}}{\sim} \text{Laplace}(0, \sigma/\lambda), \quad j = 1, \dots, P$$

$$\lambda^2 \sim \text{Gamma}(2, 1),$$

$$\sigma \sim \text{HalfNormal}(1),$$

$$\alpha \sim \mathcal{N}(0, 10^2),$$

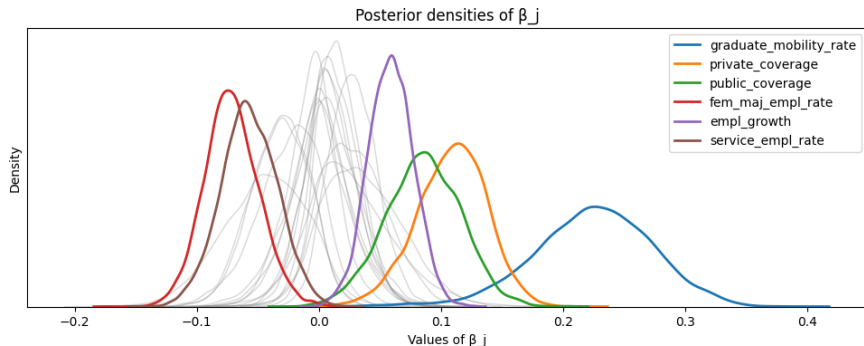
$$\boldsymbol{\phi} \mid \tau_c, \rho, W \sim \mathcal{N}_N(\mathbf{0}, [\tau_c(D - \rho W)]^{-1}),$$

$$\rho \sim \text{Beta}(2, 2),$$

$$\tau_c \sim \text{Gamma}(1, 1).$$

Selection criterion : 95% credible interval does not contain 0

Selection Results



Selected Variables

- ECEC variables: private_coverage, public_coverage
- Non-ECEC variables: fem_maj_empl_rate, service_empl_rate, graduate_mobility_rate, empl_growth

Sparse Finite Mixture Model

$$y_i^* \mid \alpha, \boldsymbol{\pi}, \{\boldsymbol{\beta}_k, \sigma_k^2\}_{k=1}^K, \phi_i \sim \sum_{k=1}^K \pi_k \mathcal{N}\left(\alpha + \mathbf{x}_i^{*\top} \boldsymbol{\beta}_k + \phi_i, \sigma_k^2\right), \quad i = 1, \dots, N$$

$$\boldsymbol{\pi} \sim \text{Dirichlet}(0.5 \mathbf{1}_K),$$

$$\alpha \sim \mathcal{N}(0, 2),$$

$$\boldsymbol{\beta}_{j,k} \mid b_{j,k}, \lambda_j \stackrel{\text{ind}}{\sim} \mathcal{N}(b_{j,k}, \lambda_j), \quad j = 1, \dots, P, \quad k = 1, \dots, K$$

$$b_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, P$$

$$\lambda_j \stackrel{\text{iid}}{\sim} \text{Gamma}(0.5, 0.5), \quad j = 1, \dots, P$$

$$\sigma_k^2 = \frac{1}{\nu_k}, \quad \nu_k \stackrel{\text{iid}}{\sim} \text{Gamma}(2, 1), \quad k = 1, \dots, K$$

$$\phi_i = \phi_{\text{raw},i} - \frac{1}{N} \sum_{i'=1}^N \phi_{\text{raw},i'}, \quad i = 1, \dots, N$$

$$\boldsymbol{\phi}_{\text{raw}} \mid \tau, \rho, W \sim \mathcal{N}_N\left(\mathbf{0}, [\tau\{\rho(D - W) + (1 - \rho)I_N\}]^{-1}\right),$$

$$\tau \sim \text{Gamma}(2, 1),$$

$$\rho \sim \text{Beta}(1, 1).$$

Label Assignment

- **Latent labels:** for each province i , $z_i \in \{1, \dots, K\}$.
- **Component mean:** $\mu_{ik} = \alpha + x_i^{*\top} \beta_k + \phi_i$.
- **Allocation weights:** $w_{ik} = \pi_k \mathcal{N}(y_i^* | \mu_{ik}, \sigma_k^2)$, $r_{ik} = \frac{w_{ik}}{\sum_{h=1}^K w_{ih}}$.
- **Sampling step:** $z_i | r_i \sim \text{Categorical}(r_i)$ (one label vector per posterior draw).
- **Posterior Similarity Matrix (PSM):**
$$S_{ii'} = \Pr(z_i = z_{i'} | \text{data}) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{I}(z_i^{(m)} = z_{i'}^{(m)}).$$
- **Point estimate of the partition (Binder loss):** choose $\hat{z}$ minimizing
$$\sum_{i < i'} [a S_{ii'} \mathbb{I}(\hat{z}_i \neq \hat{z}_{i'}) + b (1 - S_{ii'}) \mathbb{I}(\hat{z}_i = \hat{z}_{i'})].$$
- **Output:** $\hat{z}$ gives the final cluster labels used for interpretation.

Estimate cluster-specific coefficients

$$y_i^* \mid \alpha, \beta_{c_i}, \phi_i, \sigma_{c_i} \stackrel{\text{ind}}{\sim} \mathcal{N}\left(\alpha + \mathbf{x}_i^{*\top} \beta_{c_i} + \phi_i, \sigma_{c_i}^2\right), \quad i = 1, \dots, N$$

$$\alpha \sim \mathcal{N}(0, 2),$$

$$\beta_{j,c} \mid b_{j,c}, \lambda_j \stackrel{\text{ind}}{\sim} \mathcal{N}\left(b_{j,c}, \lambda_j\right), \quad j = 1, \dots, P, \quad c = 1, \dots, C$$

$$b_{j,c} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, P, \quad c = 1, \dots, C$$

$$\lambda_j \stackrel{\text{iid}}{\sim} \text{Gamma}(0.5, 0.5), \quad j = 1, \dots, P$$

$$\sigma_c^2 = \frac{1}{\nu_c}, \quad \nu_c \stackrel{\text{iid}}{\sim} \text{Gamma}(2, 1), \quad c = 1, \dots, C$$

$$\phi_i = \phi_{\text{raw},i} - \frac{1}{N} \sum_{i'=1}^N \phi_{\text{raw},i'}, \quad i = 1, \dots, N$$

$$\phi_{\text{raw}} \mid \tau, \rho, W \sim \mathcal{N}_N\left(\mathbf{0}, \left[\tau\{\rho(D - W) + (1 - \rho)I_N\}\right]^{-1}\right),$$

$$\tau \sim \text{Gamma}(2, 1),$$

$$\rho \sim \text{Beta}(1, 1).$$

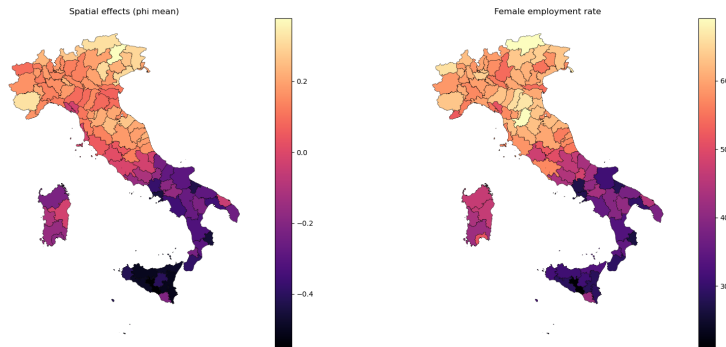
Results

- 1 cluster recovered after the label assignment step.
- Posterior summaries for intercept and regression coefficients after fitting the model with $C = 1$:

Parameter	Mean	SD	5%	50%	95%
Intercept (α)	0.053	0.021	0.020	0.053	0.087
private_coverage	0.104	0.037	0.044	0.105	0.165
fem_maj_empl_rate	-0.080	0.031	-0.130	-0.081	-0.029
public_coverage	0.151	0.034	0.095	0.150	0.206
service_empl_rate	-0.059	0.031	-0.110	-0.059	-0.008
graduate_mobility_rate	0.254	0.046	0.177	0.255	0.327
empl_growth	0.051	0.027	0.007	0.051	0.094

Results

- Posterior mean of the spatial random effects (left) and Observed female employment rate by province (right):



Conclusion

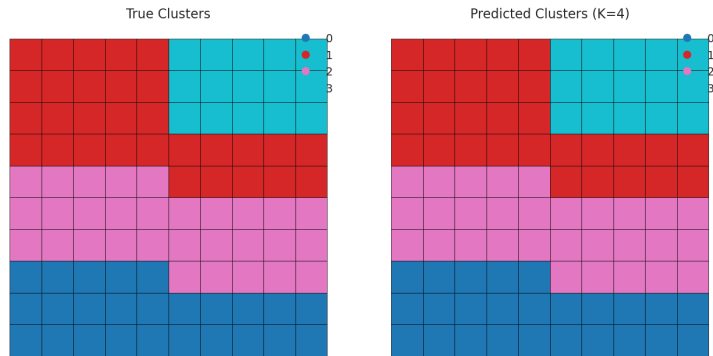
- Single cluster identified: no evidence of heterogeneous regression effects across provinces.
- High spatial correlation.
- Public and private coverage positively affect the female employment rate.
- Possible extension: multi-year data for spatio-temporal modeling.

Simulation Study I

Data generation:

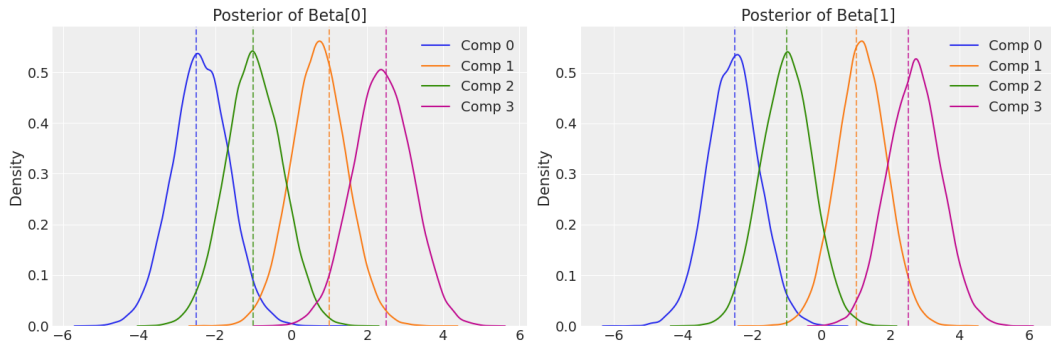
- Observations: 100 units on a 10×10 grid (adjacent if sharing a boundary).
- Spatial effect: ϕ_i simulated with $\tau = 1, \rho = 0.9$.
- Covariates: $x_i \in \mathbb{R}^2$ i.i.d. Gaussian.
- Fixed cluster labels and cluster specific regression coefficients β_k .
- Outcome: $y_i = \phi_i + x_i^\top \beta_k + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
- Experiments: $K_0 = 4$ and $K_0 = 7$.

Simulation Study II



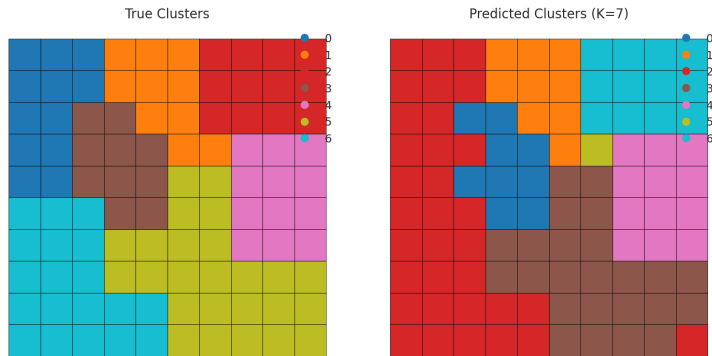
Comparison between true cluster label and predicted cluster label for the model
with $K_0 = 4$

Simulation Study III



Posterior density of the cluster-specific regression coefficients β and their true value for the $K_0 = 4$ configuration.

Simulation Study IV



Comparison between true cluster label and predicted cluster label for the model
with $K_0 = 7$

Appendix - Variables Details I

Highlighted rows show computed variables, not directly taken.

Variable	Description
fem_empl_rate	Employed women (15–64) / total women (15–64)
population	Residents on January 1st
unempl_rate	Unemployed (15–64) / labor force (15–64)
part_rate	Labor force (15–89) / population (15–89)
empl_rate	Employed (15–64) / labor force (15–64)
empl_growth	Annual growth of employed population
inact_rate	Inactive (15–64) / population (15–64)
fem_edu_rate	Women 20+ with tertiary education
male_edu_rate	Men 20+ with tertiary education
service_empl_rate	Workers in services (15–64) / all workers
GDP	Gross Domestic Product at market prices per person
n_large_comp	Local units with 250+ employees / active population (in thousand)
n_self_empl	Number of self-employed / active population
young_neet	15–29 not in employment, education or training / 15–29 people

Appendix - Variables Details II

Variable	Description
fem_house_rate	Inactive women / children aged 0–2
foreign_rate	Foreigners / total population
n_members_family	Average number of members per family
elderly_in_care	65+ in home or integrated care services
house_price	Average price of housing (€/sqm)
fertility_rate	Live births per 1,000 residents
first_birth_maternal_age	Average age at first childbirth
avg_children_per_woman	Total fertility rate (TFR)
dependency_rate	Non-working / working-age population
ageing_index	65+ / 0–14 population
net_migration_rate	Net migration per 1,000 residents
adj_net_migration_rate	Migration adjusted for natural change
area_km2	Total surface area

Appendix - Variables Details III

Variable	Description
public_transport_supply	Public transport supply per capita
internet_coverage_rate	% of households with high-speed internet
graduate_mobility_rate	Net migration of graduates (25–39)
green_m2_capita	Green space per capita in provincial capitals
eco_urban_index	Environmental performance score
per_capita_public_expenditure	Municipal spending per 0–2 year-old
ecec_participation	% of 0–2 y.o. using early childhood services
coverage*	Number of available childcare services / children aged 0-2
urban_type	1: Urban, 2: Intermediate, 3: Rural
per_capita_user_contribution	Per-child user spending (€ per 0–2 y.o.)
advanced_business_empl	Employees in advanced business services per active population
lifelong_learn	% of population aged 25–64 in lifelong learning
startup_ratio	Innovative startups per 1,000 corporations
fem_maj_empl_rate	Workers in Education and Health and Social Care / all workers

References I

[Lee(2013)] Lee, D. (2013).

CARBayes: An R Package for Bayesian Spatial Modeling with Conditional Autoregressive Priors.

Journal of Statistical Software, 55(13), 1–24.

<http://doi.org/10.18637/JSS.V055.I13>

[Park and Casella(2008)] Park, T. and Casella, G. (2008).

The Bayesian Lasso.

Journal of the American Statistical Association, 103, 681–686.

<http://doi.org/10.1198/0162145080000000337>

[Besag(1974)] Besag, J. (1974).

Spatial Interaction and the Statistical Analysis of Lattice Systems.

Journal of the Royal Statistical Society: Series B, 36(2), 192–225.

<http://doi.org/10.1111/J.2517-6161.1974.TB00999.X>

References II

[Beraha et al.(2021)] Beraha, M., Pegoraro, M., Peli, R., and Guglielmi, A. (2021).

Spatially dependent mixture models via the logistic multivariate CAR prior.

Spatial Statistics, 46, 100548.

<http://doi.org/10.1016/J.SPASTA.2021.100548>

[Mozdzen et al.(2022)] Mozdzen, A., Cremaschi, A., Cadonna, A., Guglielmi, A., and Kastner, G. (2022).

Bayesian modeling and clustering for spatio-temporal areal data: An application to Italian unemployment.

Spatial Statistics, 52, 100715.

<http://doi.org/10.1016/J.SPASTA.2022.100715>

[Moran(1950)] Moran, P. A. P. (1950).

Notes on Continuous Stochastic Phenomena.

Biometrika, 37(1/2), 17.

<http://doi.org/10.2307/2332142>

References III

[Anselin(1995)] Anselin, L. (1995).

Local Indicators of Spatial Association—LISA.

Geographical Analysis, 27(2), 93–115.

<http://doi.org/10.1111/J.1538-4632.1995.TB00338.X>

[Malsiner-Walli et al.(2014)] Malsiner-Walli, G., Frühwirth-Schnatter, S., and Grün, B. (2014).

Model-based clustering based on sparse finite Gaussian mixtures.

Statistics and Computing, 26(1), 303.

<http://doi.org/10.1007/S11222-014-9500-2>

[Leroux et al.(2000)] Leroux, B. G., Lei, X., and Breslow, N. (2000).

Estimation of Disease Rates in Small Areas: A new Mixed Model for Spatial Dependence.

In M. E. Halloran and D. Berry (eds.),

Statistical Models in Epidemiology, the Environment, and Clinical Trials, pp. 179–191. Springer.

https://doi.org/10.1007/978-1-4612-1284-3_4

References IV

[Binder(1978)] Binder, D. A. (1978).

Bayesian cluster analysis.

Biometrika, 65(1), 31–38.

<https://doi.org/10.1093/biomet/65.1.31>

[Duncan et al.(2024)] Duncan, E. W., Cramb, S. M., Baade, P. D., Mengersen, K. L., Saunders, T., and Aitken, J. F. (2024).

Developing a Cancer Atlas using Bayesian Methods: A Practical Guide for Application and Interpretation (2nd ed.).

Queensland University of Technology (QUT) and Cancer Council Queensland, Brisbane.

<https://atlas.cancer.org.au/ebook/ebook2/Index.html>